Data Analysis of Berkeley's Street Safety from Crowdsourced Collision and Pothole Reports

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Group 5

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Track A

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Problem & Motivation:

As Berkeley students, transportation is a key aspect of daily student life since housing is all off the campus. Whether people walk, bike, scooter, or drive to campus, getting to the destination safely is extremely important. However, because of so many people sharing the road with different modes of transportation, danger can arise. Furthermore, with how much use the streets get combined with how old the city is, streets can suffer from problems like potholes and cracks, which further lead to danger on the roads. As e-scooter riders, the authors of this report have both experienced our fair share of both nice roads with protected bike lanes near campus, as well as more dangerous roads with large potholes and cracks, (in which swerving around to avoid them will cause more danger from oncoming vehicle traffic). This was our motivation behind the project.

This research will impact the people who use the streets of the city of Berkeley, including drivers, cyclists, e-scooter riders, and pedestrians. Streets with potholes or cracks in need of repair affect everyone who uses the streets, not just students near UC Berkeley. This also involves city officials and workers responsible for fixing the roads.

For this report, we will be identifying the specific streets that have the most collision reports, as reported by the citizens of Berkeley. We will then correlate these with the pothole reports to determine which streets are in most need of attention and repair. By identifying specific causes for average higher collision rates and correlating it with pothole reports, we can determine which streets need the most repair, and perhaps recommend remedies for that.

Updated Research Question/Hypothesis

Previous Research Question:

Between 2020 and 2025, how have reported potholes in Berkeley affected the safety and mobility of drivers, cyclists, e-scooter riders, and pedestrians, and which areas show the highest concentrations of hazards?

New Research Question:

Which streets in Berkeley, CA require the most repair during the period 2020–2025, and is there any correlation between the number of reported potholes and incident frequency among drivers, cyclists, e-scooter riders, and pedestrians?

Measurements/Metrics

- Streets in need of repair = most reports of potholes, cracks

- Measuring number of incidents/collisions with anyone using the roads - drivers, cyclists, e-scooter riders, and pedestrians
- Measuring number of pothole reports

Datasets

Collisions:

- Street Story Collision Incidents
    - CSV sheet (raw): Street Story Reports - BERKELEY (City) - 10_30_2025
        - Cleaned data: second sheet in above (in ‘Cleaned_Data’ sheet)
    - Coverage: City of Berkeley, Oct 2018–Oct 2025
    - Granularity: Spatial data, street-level, each row is an incident report
    - Type: Categorical
    - Format: Tabular, CSV
    - Source: Street Story, crowdsourced

Pothole Reports:

- SeeClickFix - City of Berkeley
- Berkeley Service Report (Potholes) Dataset
    - Go to Actions → Query Data
    - CSV sheet (raw): 311_Cases_-_COB_20251109 - Potholes
        - Cleaned data: second sheet in above (in ‘Cleaned_Data’ sheet)
    - 770k rows, 7.2k pothole reports
    - Coverage: City of Berkeley, Aug 2010 –Nov 2025
    - Granularity: Spatial data, street-level, each row is a case
    - Type: Categorical
    - Format: Tabular, CSV
    - Source: City of Berkley Open Data Portal

Data & Cleaning:

The Street Story Reports Dataset contains information regarding traffic-pedestrian related collisions from reports filed from residents living in Berkeley. The coverage of this dataset is the City of Berkeley from Oct 2018 to Oct 2025. Each row is an incident report, contains categorical data, and is spatial, with appended latitude and longitude coordinates. The format of this data is a tabular, CSV file and is sourced from Street Story, an open crowdsourced platform allowing users to anonymously report collision data for city and public use and analysis.

In the data cleaning process, we noticed that some of the rows were missing addresses, but still contained latitude and longitude coordinates. We also found that the date of a crash or near miss was inconsistently formatted, with some being formatted as 'MM/DD/YYYY' and others being 'MM/YYYY'. To remedy this, we simply created new columns containing just the month and year by parsing the column using Regex. Regarding the data itself, we also found that the severity of each incident was not weighted the same, and it depended on the narrative how serious the report was. For example, some reports indicated severe collisions with a motor vehicle, while others simply noted that drivers in the area were simply more aggressive, not taking note of the physical street hazard. In the coordinates column, we noticed that the types of the coordinates were inconsistent, with some being polylines (representing streets or blocks) instead of points, which were represented by an array of coordinates. We solved this by taking the 2 middle coordinates of the array and taking the average to get the middle of a street.

Regarding missing data in this dataset, we can use the missing data types in the Dark Data book to categorize the types of data that are unseen and purposely omitted. For example, Dark Data Type 4 (self-selection) is apparent in this dataset due to the data being self-reported and voluntary. Only people with a motivation to self-report collisions do report them, and there is the possibility of under or over-reporting incidents. We also notice that there is missing data due to location and proximity to campus, or Dark Data Type 3 (selection bias). Most of the data points can be aggregated at UC Berkeley campus's adjacent streets, with fewer points as the proximity to campus becomes greater. In order to counteract these discrepancies, we decided to aggregate collisions or near misses from incidents that are on streets immediately adjacent to the campus to avoid reporting bias by essentially grouping points that are very proximal to each other.

The SeeClickFix Pothole Reports Dataset contains information regarding pothole incidents from reports filed from residents living in Berkeley. This dataset contains over 770k cases, with around 7.2k of those cases being pothole reports. The coverage of this dataset is the City of Berkeley from Aug 2010 to Nov 2025. Each row is a case, contains categorical data, and is spatial, with appended latitude and longitude coordinates. The format of this data is a tabular, CSV file and is sourced from the City of Berkeley Open Data Portal, a city-managed open-source data repository containing data related to the City of Berkeley that can be used by the public.

Initial Results & Visualizations:

Working Notebook:

Group_5_Final_Project_Data_Analysis.ipynb

Maps and charts are located in the /visualizations folder.

From the initial visualizations, it appears that the majority of reports for both collisions and pothole reports are centered around UC Berkeley campus and that there are natural clusters of data points in more busy streets (such as Telegraph Ave.). Some caveats are that there are a few outlier data points, such as some reports being outside of City of Berkeley land boundaries and that the data from both datasets does not match the same period of time, since the pothole reports dates back to 2010.

Progress Since Proposal

Since we've turned in our proposal, we have changed our research question. We also narrowed down our data, and figured out that lots of what we found last time was unusable. We then discovered the two datasets we are currently using, and cleaned it up.

In our proposal, by this midterm we wanted to "Analyze and put together data for potholes and street conditions", "Gather data for accident location and types", and "Analyze locations of data." We were able to do this, as well as put together maps that visualize the different datasets, which we had wanted to have done by the final due date.

Plan to Completion

In order to complete our project, we need to analyze the data together and see if there truly are correlations between the incidents on the road, and the pothole reports. We also need to finalize the cleanup of our data, and have our interactive maps ready. Additionally, we need to create and finalize our ArcGIS StoryMap. Given that we have RRR week to work on this, theoretically the final deliverables will all be finished in time by the due date.

References

City of Berkeley. “311 Cases - COB.” *Cityofberkeley.info*, 16 Aug. 2017, data.cityofberkeley.info/311/311-Cases-COB/bscu-qpbu/about_data. Accessed 11 Nov. 2025.

“SeeClickFix - City of Berkeley.” *Seeclickfix.com*, 2025, seeclickfix.com/web_portal/Q4nTBJPnrfGyosn85v3Js1Uq/issues/map?lat=37.860208869965135&lng=-122.25353673912637&max_lat=37.87235460732473&max_lng=-122.23276571251509&min_lat=37.848061130803465&min_lng=-122.27430776573769&search=pothole&status=open%2Cacknowledged%2Cclosed%2Carchived&zoom=14. Accessed 11 Nov. 2025.

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